ASIRIVE // CORTEX

TKKYIA Submission //
Hybrid Edge-Server Wearable

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Navigation
INDEPENDENCE

Made by Haziq, Irfan, Eryna

Designed with SAVH Singapore

https://www.asirive.com

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INTRODUCTION

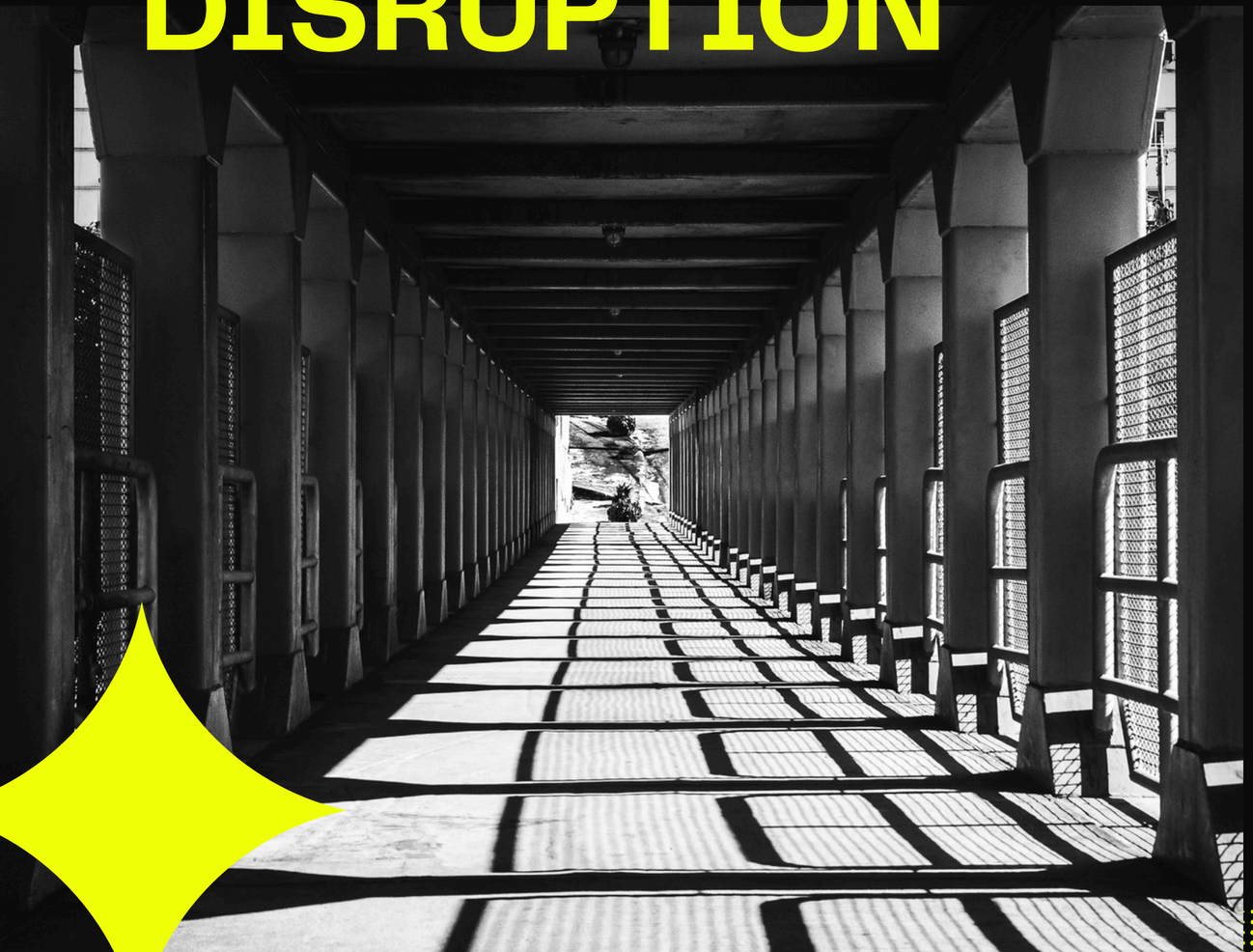
The **assistive tech market** forces the visually impaired to **choose** between their **white cane** and a **smartphone**. Asirive Cortex **shatters** this bottleneck. We engineered a hands-free, distributed wearable that processes physical reality at the **speed the user walks**. By **separating** local edge-compute (for instant physical hazard detection) from cloud-based reasoning, Cortex delivers true navigation independence **without the premium price tag**

Architecture Hybrid Edge-Server (Raspberry Pi 5 + Hailo-8L + Gemini Live)

Economics Sub-$250 BOM. Breaking the $3,000 premium medical monopoly.

Validation Live-validated with SAVH in GPS-denied environments

THE ASIRIVE CORTEX DISRUPTION



DISRUPTIVE UNIT ECONOMICS

Traditional assistive wearables are locked behind a $1000 – $3000 monopoly. By utilizing off-the-shelf commodity hardware and a modular software stack, Asirive achieves a 90% cost reduction (SGD 250 BOM). This allows us to scale rapidly into the medical market without the high manufacturing barriers of custom silicon

HYBRID ARCHITECTURAL RESILIENCE

Most startups fail by relying 100% on cloud AI. Asirive uses a 'Layered Safety' model. Immediate hazard detection runs locally on the edge (Hailo-8L), while complex reasoning is pushed to the cloud (Gemini). This ensures the product remains functional in high-density, GPS-dead zones like VivoCity or MRT stations where competitors fail

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HYBRID ORCHESTRATION

Without GPS



THE CLOUD BRAIN

Gemini Live acts as the cognitive agent. It uses real-time function calling to understand the surroundings, read overhead signs, and logically route the user through complex environments

LOCAL REFLEXES

Simultaneously, the depth and yolo model runs entirely offline. Acting as the system's reflexes, it processes raw visual frames to detect immediate physical hazards—like sudden drop-offs or moving crowds— with low latency

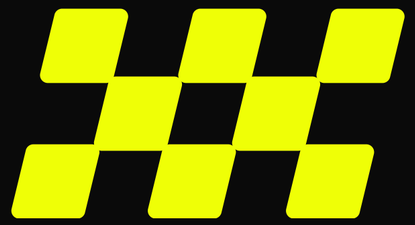
ARCHITECTURAL RESILIENCE

Intent-routing synchronizes both streams. If cellular networks crash in a dense crowd, the cloud brain gracefully pauses, but the local reflexes never stop. The user loses directions, but never loses their physical safety net

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THE TECHNOLOGY STACK



EDGE PERCEPTION

Raspberry Pi 5 paired with a Hailo-8L NPU. Runs Ultralytics YOLO and FastDepth locally to process raw camera frames, dropping sub-100ms physical hazard alerts entirely offline for baseline structural safety

CLOUD COGNITIVE

Powered by Google Gemini 3.1 Flash Live. Handles multimodal intent routing, complex spatial understanding, and API function calling (like live transit telemetry) when the user requires deep environmental context

DETERMINISTIC AUDIO PIPELINE

Utilizes Cartesia and Kokoro TTS models. Delivers low-latency, conversational vocal feedback directly to the user's earpiece without blocking the ambient environmental noise crucial for their mobility

FAILING WITH HONOUR // ARCHITECTURAL BOTTLENECKS»

ARROGANT DESIGN ASSUMPTIONS

Initially, we built a complex 3D spatial audio system. Live SAVH testing proved it was a total failure—the spatial audio was overwhelming and blocked crucial ambient traffic noise. We scrapped the engine entirely to deliver concise, targeted commands, embracing the cane instead of trying to replace it

THERMAL THROTTLING & DEPENDENCY HELL

Attempting to stream continuous raw video and audio simultaneously to the cloud choked the Raspberry Pi 5. The thermal load caused massive latency and the constant read/write cycles corrupted three separate SD cards during testing

THE ASYNC PIPELINE OVERHAUL

We solved this with a brutal architectural overhaul. We rewrote the async pipeline to throttle video inference to 1–2 frames per second while keeping the audio continuous. This instantly killed the latency bottleneck, stopped the overheating, and birthed our hybrid edge-server architecture



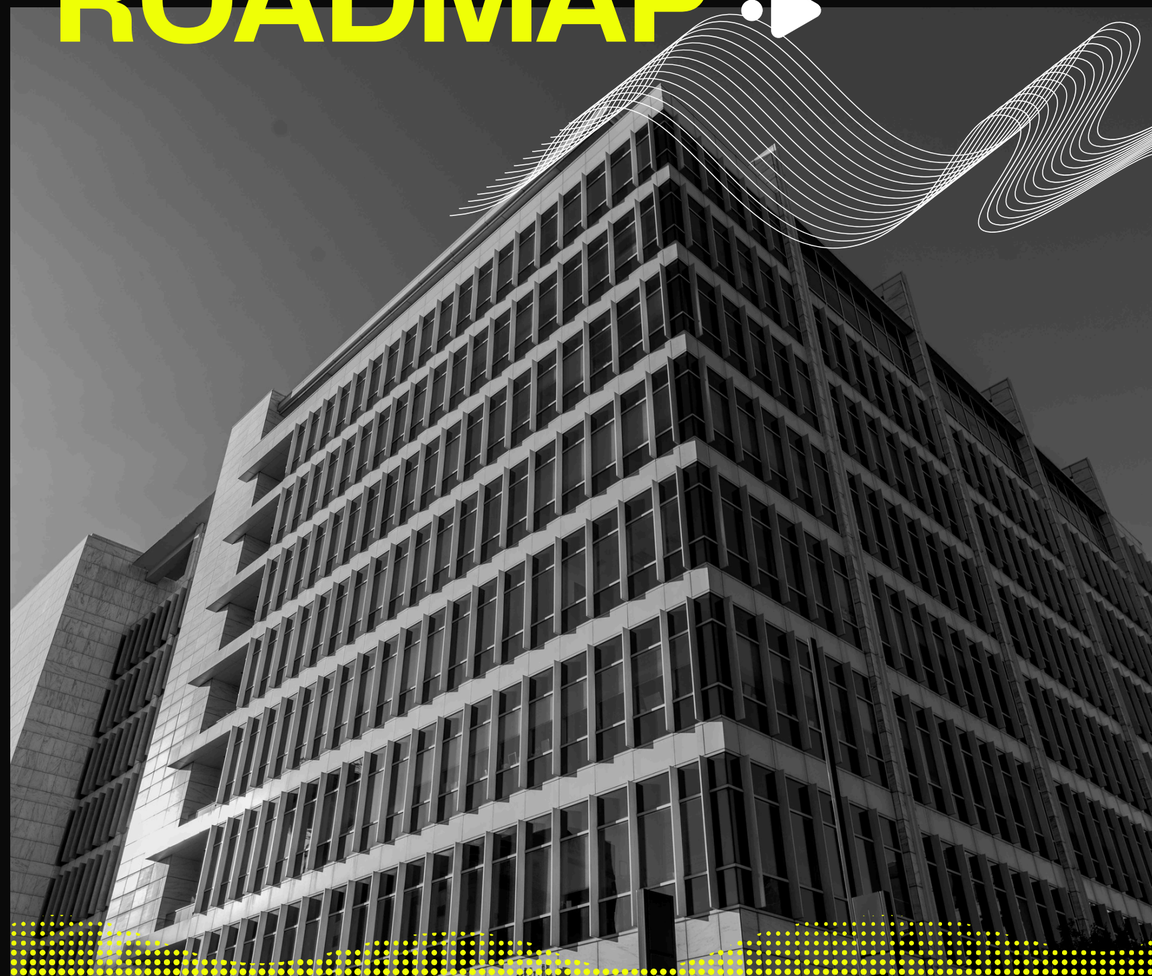
THE ENGINEERING ROADMAP

PHASE 1: HARDWARE MINIATURIZATIO N

The current off-the-shelf Raspberry Pi 5 is a functional lab prototype, but too bulky for daily mobility. We are fabricating a custom Single Board Computer (SBC) to remove I/O ports, optimize thermal management, and hit a 12-hour continuous battery life target

PHASE 2: EDGE-NATIVE ECOSYSTEM

To guarantee stability in network dead zones, we are transitioning from cloud-streaming to an entirely local edge-native audio model. Concurrently, we are developing a Caregiver App for remote monitoring, an open SDK for localized smart-city transit intergrations and the custom model



THE ASIRIVE TEAM →

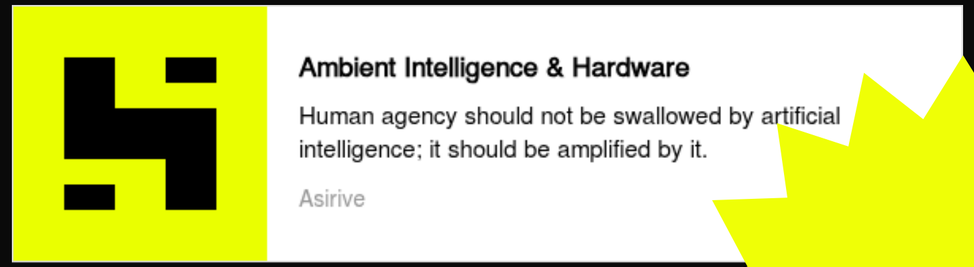
HAZIQ	Systems Architect & Founder. Engineered the dual-track edge-server architecture, YOLO intent-routing, and asynchronous Python pipelines
IRFAN	3d Designer & Validation. Facilitated physical SAVH deployments, live environmental stress-testing, and operational logistics
ERYNA	Hardware Assembly & Data Operations. Managed the $250 BOM unit economics and latency data collection during live SAVH field trials

Breaking the expensive monopoly with hybrid edge-server architecture

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THANK YOU! »»





https://youtu.be/OENEq4wxqr8

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